

WHERE DATA BECOMES CONSCIOUS

Note

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This document is for discussion purposes only. Design is subject to change.



P R O J E C T 4

Large Scale Attic Layout Collision Prevention System

Sensor-Based vs Camera-Based Approach Analysis — 100m Track / 3 Trains

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IMPORTANT NOTICE — DRAFT DOCUMENT

This document is a PRELIMINARY PROPOSAL only. It is intended to outline possible technical approaches, answer Harry's initial questions, and provide a basis for further discussion. No designs, specifications, or cost estimates in this document are final. All details are subject to change following:

- Site visit and physical inspection of the attic space
- Confirmation of the final track plan and dimensions
- Discussion of budget and preferred technology direction
- Testing and evaluation of approach on a small section first

! Do not purchase any hardware based solely on this document. Hardware recommendations are indicative only. A finalised specification will be provided before any purchase is made.

1. Executive Summary

Harry is planning a significantly larger model train layout — approximately 100 metres of total track in an attic space, with a main loop of around 20 metres. The goal is to run 2 to 3 trains simultaneously on this layout without collisions. This document evaluates two distinct technical approaches and answers each of Harry's specific questions.

Layout at a Glance

Location	Attic (new layout)
Main line length	~20 metres
Total track length	~100 metres
Track complexity	High — complex loops, multiple sidings
Simultaneous trains	2 to 3 maximum
Train type	LionChief — Bluetooth (BLE)
Target completion	August 2026 (track laying)

Two Approaches Evaluated

Option A — Sensor-Based	Option B — Camera-Based
Reed switches at block boundaries ESP32 microcontroller brain 100% reliable, no light dependency Low cost, proven technology Recommended for this layout	Multiple overhead cameras Dedicated PC for processing Visual tracking of train positions Higher cost, complex setup Feasible but not recommended

2. Harry's Questions — Answered

Q1: How many sections and sensors would make sense?

The track should be divided into blocks using a concept called Block Signaling — the same approach used in real railways. Each block needs one sensor at its entry point. A train entering a block triggers the sensor; the system then knows that block is occupied.

Rule of thumb: you need at least 3x the number of trains in blocks so trains always have empty space between them.

Trains Running	Minimum Blocks	Recommended Blocks
2 trains	6	10–12
3 trains	9	12–15

i For Harry's layout: 12 to 15 blocks recommended. At 100m total track ÷ 12 blocks = ~8m per block average. Each block needs 1 reed switch sensor = 12 to 15 sensors total.

Q2: Which sensor type is recommended?

Three sensor types are used in model railroads. Reed switches are the clear winner for this application:

Sensor Type	Cost	Reliability	Notes
Reed Switch	\$0.50–\$1 each	100% — mechanical	Magnet under loco. No track wiring. Works anywhere.
IR Sensor	\$1–\$3 each	Good — optical	Affected by ambient light. Needs alignment under track.
Hall Effect	\$1–\$2 each	Very good	Similar to reed switch. Slightly more complex wiring.

i RECOMMENDATION: Reed switches (model MC-38). A small magnet is glued under each locomotive. When the train passes the sensor, the magnet closes the reed switch and the ESP32 detects it instantly.

Q3: Dedicated PC or ESP32 microcontroller?

For the sensor-based approach, an ESP32 microcontroller is completely sufficient and is in fact the better choice. No PC is required.

ESP32 Microcontroller	Dedicated PC
Cost: \$5–8 Built-in WiFi for web dashboard 15+ GPIO pins for sensors BLE to control trains directly Runs 24/7 on USB power bank No screen, no keyboard needed	Cost: \$300–600 Required ONLY for camera approach Overkill for sensor logic Needs display and peripherals Higher power consumption More complex to maintain

i RECOMMENDATION: One ESP32 as the master controller. It reads all reed switches, runs block logic, sends BLE commands to trains, and serves the web dashboard — all in one \$8 device.

Q4: Will Bluetooth (BLE) work reliably at 20m in an attic?

BLE range indoors is typically 10–15 metres. A 20-metre attic with metal beams and interference makes reliable BLE coverage across the full layout uncertain. Three strategies are available:

Strategy 1	Place ESP32 in the centre of the layout (~10m to each end). Most cases this will work.
Strategy 2	Use two ESP32 units — one at each end — and coordinate them via WiFi.
Strategy 3	If buying new locomotives for this layout, switch to DCC. Signal runs through the track rails — unlimited range, industry standard for large layouts.

! DRAFT NOTE: BLE range must be tested physically in the attic before finalising this approach. Strategy will be decided after on-site testing.

3. Option A — Sensor-Based System (Recommended)

This approach divides the track into blocks using reed switch sensors. The ESP32 always knows which blocks are occupied and controls train speeds accordingly. It is the same principle used in real railway signaling.

How It Works — Step by Step Workflow

1	Train moves	A magnet under the locomotive approaches a reed switch at a block boundary
2	Sensor fires	Reed switch closes → GPIO pin on ESP32 goes HIGH → block marked OCCUPIED
3	Block logic runs	ESP32 checks: is the NEXT block occupied? Is the block after that occupied?
4	Speed decision	If next block clear → maintain speed. If next block occupied → slow down. If same block → emergency stop.
5	BLE command	ESP32 sends speed command via Bluetooth directly to the affected LionChief locomotive
6	Train responds	Locomotive adjusts speed within 150ms of receiving command
7	Block clears	When train exits a block, the EXIT sensor fires → block marked CLEAR → following train can resume
8	Dashboard	Web page on any phone shows live block map — green = clear, red = occupied

Block Map — What the System Tracks

The layout is divided into named blocks. Each block has an entry sensor. The system maintains a live map like this:

B01	B02	B03	B04	B05	B06	B07	B08	B09	B10
CLEAR	TRAIN 1	CLEAR	CLEAR	TRAIN 2	CLEAR	CLEAR	TRAIN 3	CLEAR	CLEAR

Red = occupied by a train | Green = clear | System automatically slows any train approaching a red block.

Hardware Required

ESP32 dev board	1 unit, master controller, WiFi server, BLE hub
Reed switches (MC-38)	15–18 units, one per block entry point
Small round magnets	One per locomotive (glued under body)
22AWG wire	50–80 metres to wire sensors back to ESP32
USB power supply	5V 2A, to power ESP32 permanently
Junction box	To house ESP32 and wiring neatly

i IMPORTANT: Reed switches must be installed DURING track laying in September — much easier than retrofitting after track is fixed. Coordinate sensor placement with track installation.

Advantages of Sensor-Based Approach

- 100% reliable: no cameras, no lighting dependency, no vision algorithms
- Works in any light level — attic lighting irrelevant
- Works at any track size — add more sensors as layout grows
- Very low cost — under \$50 total hardware
- Proven technology — used in real model railroad clubs worldwide
- No PC required — ESP32 handles everything

Limitations

- Sensors must be installed during track laying — harder to retrofit
- Does not know WHERE on the track the train is between sensors
- Block resolution limited to sensor placement spacing

4. Option B: Camera-Based System

This approach uses overhead cameras to visually track train positions in real time, similar to what was built in Projects 2 and 3, but scaled up significantly for the large attic layout.

! DRAFT NOTE: The camera approach was the method used in Projects 2 and 3 on the small oval layout. While it worked as a proof of concept, it had reliability challenges even on a small layout. Scaling it to 100m of complex track would require significant investment in hardware and engineering.

How It Would Work, Workflow

1	Cameras capture	3–5 ceiling cameras positioned above different sections of the layout capture continuous video at 30fps
2	Frame stitching	PC software stitches camera feeds into one unified overhead view of the full layout
3	Train detection	Computer vision (MOG2 or ArUco markers) locates each train in the unified frame
4	Gap calculation	Software measures pixel distances between trains in each section of track
5	Speed decision	Gap-to-speed algorithm (PD controller) determines required speed for each train
6	BLE command	PC sends Bluetooth speed commands to each locomotive
7	Dashboard	Web browser shows unified camera view with tracking circles and speed controls

Challenges at 100m Scale

Challenge	Why it is difficult
Multiple cameras needed	One camera cannot cover 100m of complex track. 3–5 cameras needed, each requiring its own USB connection to the PC.
Camera stitching	Combining 5 camera feeds into one unified image requires significant software and calibration work.
Train identification	At small scale, trains look similar from above. At 100m scale this problem is worse. ArUco markers on each locomotive would help.
BLE range	PC would need to maintain BLE connections to 3 trains across a 20m space with potential interference.
Processing power	Running 5 camera streams simultaneously with real-time CV processing requires a capable dedicated PC.
Lighting consistency	Attic lighting is uneven. Poor lighting causes false detections. Consistent overhead LED lighting would be required.
Cost	3–5 cameras + capable PC + LED lighting = \$500–1000+ investment.

Hardware Required for Camera Approach

Dedicated PC	High spec laptop or mini PC — Core i5, 16GB RAM minimum
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USB cameras	3–5x 4K USB cameras (wide angle)
USB hub	7-port powered USB hub for multiple cameras
LED lighting	Consistent overhead LED strips for even illumination
Camera mounts	Ceiling mounts or tripods above each section
Calibration targets	Reference markers on layout for camera alignment

When Camera Approach Makes Sense

- Layout has good fixed overhead mounting positions for cameras
- Budget allows for \$500–1000+ hardware investment
- You want visual monitoring of the layout regardless of collision prevention
- Track is already laid and sensors cannot be easily installed

5. Side-by-Side Comparison

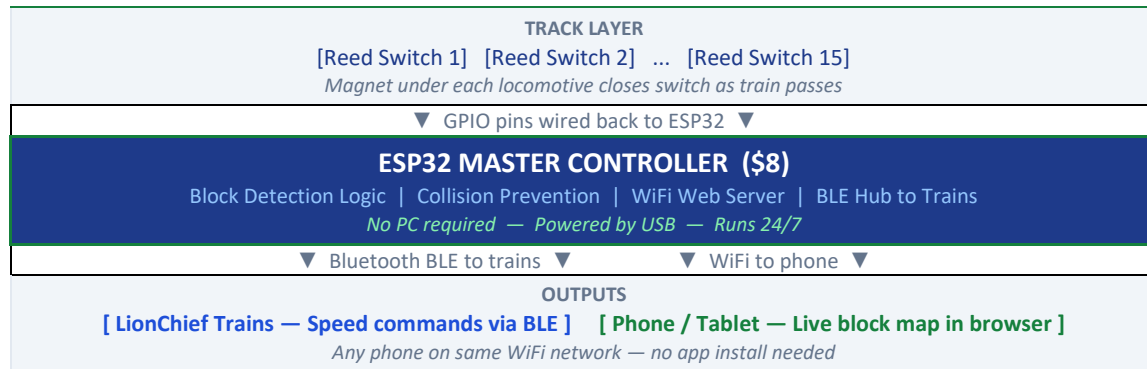
Criteria	Option A — Sensors	Option B — Camera	Best
Hardware cost	\$44 approx	\$500–1000+	S
Reliability	100% mechanical	85–90% , variable	S
Light dependency	None	High — needs good lighting	S
PC required	No ESP32 only	Yes — dedicated PC	S
Scalability	Excellent — add sensors	Hard — add cameras	S
Installation	During track laying	After track laid	S
Visual monitoring	No — block map only	Yes — see full layout	C
Setup complexity	Low	Very high	S
Maintenance	Very low	Medium–high	S
Attic suitability	Excellent	Requires good lighting	S
Future expansion	Easy	Complex	S
Proven for 100m	Yes — railway standard	Experimental at this scale	S

S	Sensor wins this criterion	C	Camera wins this criterion
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6. Recommended Architecture — Sensor System

Based on the analysis above, the sensor-based approach is recommended for the Project 4 attic layout. Below is the proposed system architecture.

System Block Diagram



Web Dashboard — What Peter Sees

The ESP32 hosts a simple web page accessible on any phone or tablet connected to the same WiFi network. No app installation needed.

Live block map	Colour-coded grid: green = clear, red = occupied, shows which block each train is in
Train speeds	Current speed of each train displayed
Manual override	Buttons to increase/decrease any train speed
Emergency stop	One button stops all trains immediately
BLE status	Shows which trains are connected
Access	http://192.168.1.xxx — any phone browser on same WiFi

7. Indicative Hardware List

! DRAFT — Do not purchase based on this list alone. Final specification will be provided after track plan is confirmed.

Sensor-Based System — Indicative Cost

Item	Qty	Unit Cost	Total	Priority
ESP32 development board (38-pin)	2	\$5	\$10	Essential
Reed switches MC-38	20	\$0.50	\$10	Essential
Small round magnets 6mm	10	\$0.50	\$5	Essential
22AWG 2-core wire (50m roll)	2	\$8	\$16	Essential
5V 2A USB power supply	1	\$8	\$8	Essential
Small plastic junction box	1	\$5	\$5	Recommended
Breadboard + jumper wires	1	\$5	\$5	Recommended
USB-C cable 2m	2	\$3	\$6	Recommended
Spare reed switches	5	\$0.50	\$3	Spare
TOTAL ESTIMATED COST			~\$68	

Proposed Timeline

Month	Phase	Key Activities
August 2026	Hardware Order	Finalise block plan from track layout. Order ESP32, reed switches, wire. Begin firmware development.
	Track Laying	Install reed switches DURING track laying. Much easier than retrofitting. Coordinate with track installation.
	Integration	Connect sensors to ESP32. Upload firmware. Test block detection on completed track.
	Testing	Multi-train testing. Tune block boundaries. Calibrate BLE range. Launch web dashboard.
	Final	Full system running. Documentation. Handover.

8. Next Steps — Discussion Required

Before this proposal can be finalised, the following points need to be discussed and confirmed between Harry and Datix AI:

1	Confirm approach: Sensor-based (recommended) or Camera-based?
2	Confirm final track plan — block boundaries need to be mapped against the green track drawing
3	Confirm number of simultaneous trains — 2 or 3? This determines number of blocks needed
4	Confirm if LionChief locomotives will be reused or if new locomotives (potentially DCC) will be purchased
5	Confirm BLE range test — ESP32 position in the attic needs to be evaluated
6	Confirm budget for Project 4 hardware
7	Confirm September track laying date — sensor installation must be coordinated with this

DRAFT — NOT FINALISED — FOR DISCUSSION ONLY

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